

WORKED EXAMPLE

Add a cost for large selected weights

The objective balances prediction error and squared weight magnitude.

$$J = \frac{1}{N} \sum_{i=1}^N \frac{1}{2} (\hat{y}_i - t_i)^2 + \frac{\lambda}{2} \sum_j w_j^2$$

N is the number of examples; lambda is the regularization coefficient. The data term is a mean. The penalty sums every selected weight once, with no extra division by N under this convention.

Lecture 5 page 137 writes the weight penalty using the squared Frobenius norm of each weight matrix: this means add the squares of all its selected entries.

Our exercise penalizes w_1 and w_2 only. The bias b is explicitly unpenalized. State the selection policy before computing the objective or gradients.

WORKED EXAMPLE

Set up the main three-parameter example

Chosen data and initial parameters; ordinary gradient descent.

$$\hat{y} = w_1 x_1 + w_2 x_2 + b$$

$$x = (2, 1), \quad t = 1, \quad (w_1, w_2, b) = (1, -2, 0.5)$$

$$\lambda = 0.2, \quad \eta = 0.1$$

$$J = \frac{1}{2}(\hat{y} - t)^2 + \frac{0.2}{2}(w_1^2 + w_2^2)$$

There is one example, so the mean data loss equals its one-example loss. Lambda controls penalty strength; eta controls the parameter update. They are different numbers with different roles.

WORKED EXAMPLE

Compute prediction and data loss

Keep the residual sign for the gradient, even though the loss squares it.

$$\hat{y} = 1(2) + (-2)(1) + 0.5 = 2 - 2 + 0.5 = 0.5$$

$$r = \hat{y} - t = 0.5 - 1 = -0.5$$

$$L_{data} = \frac{1}{2}(-0.5)^2 = \frac{1}{2}(0.25) = 0.125$$

r is prediction minus target. The negative residual says this prediction is below its target. The squared data loss is nevertheless positive.

WORKED EXAMPLE

Compute the penalty and total objective

The unpenalized bias does not appear in this sum.

$$w_1^2 + w_2^2 = 1^2 + (-2)^2 = 1 + 4 = 5$$

$$R = \frac{\lambda}{2}(w_1^2 + w_2^2) = \frac{0.2}{2}(5) = 0.5$$

$$J = L_{data} + R = 0.125 + 0.5 = 0.625$$

R denotes the regularization penalty. The negative weight contributes a positive square, so either sign can incur a large penalty.

WORKED EXAMPLE

Differentiate each part of the objective

The gradient of a sum is the sum of its gradients.

$$\frac{\partial L_{data}}{\partial w_j} = r \frac{\partial \hat{y}}{\partial w_j} = rx_j$$

$$\frac{\partial R}{\partial w_j} = \frac{\lambda}{2}(2w_j) = \lambda w_j$$

$$\frac{\partial J}{\partial w_j} = rx_j + \lambda w_j$$

$$\frac{\partial J}{\partial b} = r(1) + 0 = r$$

The factor 2 from differentiating the square cancels the penalty factor 1/2. The penalty bias derivative is zero because b was excluded.

WORKED EXAMPLE

Substitute every main gradient

All three derivatives use the original parameter state.

$$g_{w_1} = (-0.5)(2) + 0.2(1) = -1 + 0.2 = -0.8$$

$$g_{w_2} = (-0.5)(1) + 0.2(-2) = -0.5 - 0.4 = -0.9$$

$$g_b = -0.5$$

parameter	data gradient	penalty gradient	total
w1	-1	+.2	-.8
w2	-.5	-.4	-.9
b	-.5	0	-.5

The penalty gradient has the weight sign. Subtracting that contribution pulls either positive or negative weights toward zero.

WORKED EXAMPLE

Update all three parameters simultaneously

Compute every new value from the same original state.

$$w_1^{new} = 1 - 0.1(-0.8) = 1.08$$

$$w_2^{new} = -2 - 0.1(-0.9) = -1.91$$

$$b^{new} = 0.5 - 0.1(-0.5) = 0.55$$

Although arithmetic is written one parameter at a time, all gradients came from the original prediction 0.5. Do not recompute the residual halfway through.

w_1 increased overall because its data-loss contribution outweighed the penalty contribution. Regularization does not force every total update to reduce every weight magnitude.

WORKED EXAMPLE

Check the new full regularized objective

Use the new parameters in both the data term and penalty.

$$\hat{y}^{new} = 1.08(2) + (-1.91)(1) + 0.55 = 0.8$$

$$r^{new} = 0.8 - 1 = -0.2, \quad L_{data}^{new} = \frac{1}{2}(0.2)^2 = 0.02$$

$$\begin{aligned} R^{new} &= 0.1(1.08^2 + (-1.91)^2) \\ &= 0.1(1.1664 + 3.6481) = 0.48145 \end{aligned}$$

$$J^{new} = 0.02 + 0.48145 = 0.50145 < 0.625$$

This chosen step decreases the stated full objective. A different learning rate need not do so.

WORKED EXAMPLE

Derive the weight-decay form algebraically

For ordinary gradient descent on this declared L2 objective.

$$w^{new} = w - \eta(g_{data} + \lambda w)$$

$$= w - \eta g_{data} - \eta \lambda w$$

$$= (1 - \eta \lambda)w - \eta g_{data}$$

$$1 - \eta \lambda = 1 - 0.1(0.2) = 0.98$$

The shrink factor acts on the old weight; the data-gradient correction is also computed at the old state. Bias uses its ordinary data update because it is unpenalized.

WORKED EXAMPLE

Verify both weight updates using the shrink factor

The two algebraic forms must give the same result.

$$w_1^{new} = 0.98(1) - 0.1(-1) = 0.98 + 0.1 = 1.08$$

$$w_2^{new} = 0.98(-2) - 0.1(-0.5) = -1.96 + 0.05 = -1.91$$

With only the penalty and zero data gradient, the weights would become .98 and -1.96. Both magnitudes shrink. The real data gradients add the corrections shown above.

Do not use the shrink factor on b in this exercise, and do not apply both the combined gradient and an additional shrink factor: that would count the same penalty twice.

WORKED EXAMPLE

Read the boxed lecture factor carefully

Lecture page 138 shows both an expanded update and a boxed shorthand.

objective coefficient: $\lambda = 0.2$

per step decay: $d = \eta\lambda = 0.1(0.2) = 0.02$

$$w^{new} = (1 - d)w - \eta g_{data}$$

The expanded slide term is $-\eta\lambda w$. Its boxed factor $1-\lambda$ agrees only if that boxed λ is redefined as the per-step decay d ; otherwise the learning-rate factor is missing.

Keep λ fixed as the objective coefficient throughout your solution. Here $1-\lambda=.8$ would be incorrect; the required shrink factor is $1-d=.98$.

WORKED EXAMPLE

Why Adam needs a separate distinction

Ordinary gradient descent applies the same scalar rate to every gradient term.

With an L2 objective under Adam, the extra λw term enters both moment recurrences before adaptive normalization. A separately applied weight decay instead changes the weight directly.

Those operations are generally different. Do not assume that adding an L2 penalty to an adaptive optimizer is identical to multiplying weights by a decay factor outside it.

For this lesson, the exact equivalence we derived is for ordinary gradient descent with the declared mean-data-loss plus L2 objective.

WORKED EXAMPLE

Relate a scalar weight to input sensitivity

This derivative is with respect to input x , not a training gradient with respect to w .

$$f(x) = \sigma(wx), \quad \frac{df}{dx} = w\sigma(wx)(1 - \sigma(wx))$$

$$x = 0 : \quad \sigma(0) = \frac{1}{2} \implies f'(0) = w/4$$

$$w = 0.5 : \quad f'(0) = 0.5/4 = 0.125$$

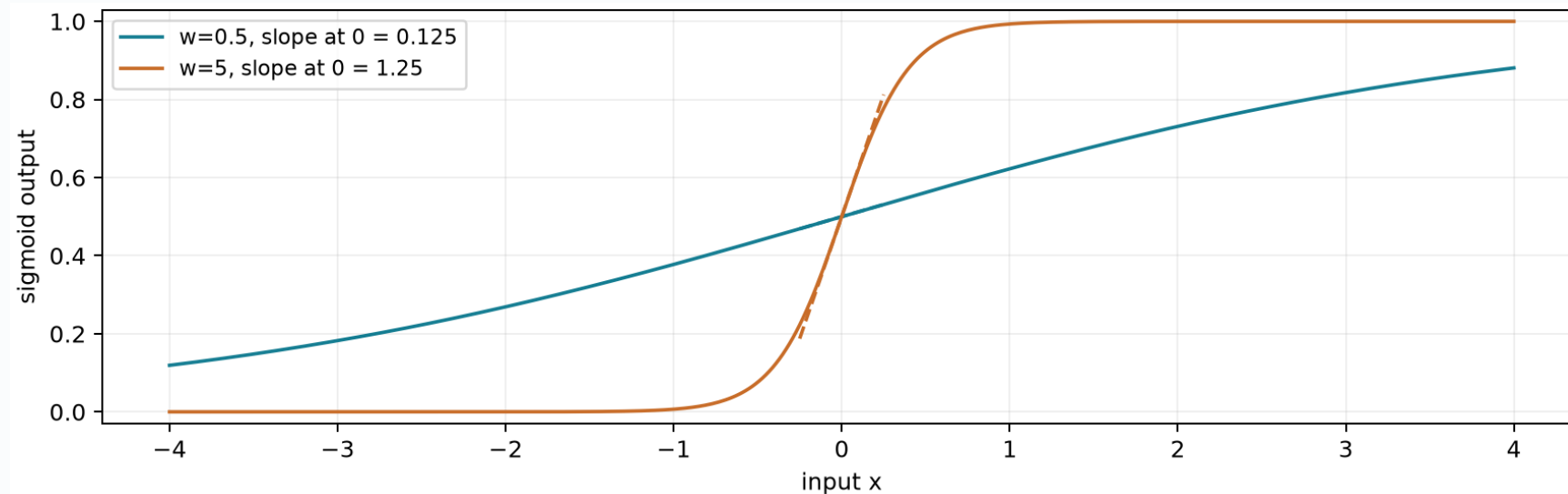
$$w = 5 : \quad f'(0) = 5/4 = 1.25$$

This supplies the arithmetic behind the scalar steepness picture on page 133. It is a local illustration, not a general theorem that every deep network with smaller weights is smoother.

WORKED EXAMPLE

See the two scalar sigmoid responses

Both pass through (0,.5), but their slopes there differ by a factor 10.



The dashed lines are local tangents at $x=0$. The steeper curve reacts more strongly to small input changes near that point.

INDEPENDENT PRACTICE

Your turn: fresh data, weights and regularization

Use the same objective convention and leave the bias unpenalized.

$$x = (1, 2), \quad t = 2, \quad (w_1, w_2, b) = (2, 1, -1)$$

$$\lambda = 0.1, \quad \eta = 0.1$$

Compute prediction, residual, data loss, penalty, total objective, all three gradients and one simultaneous update. Show the separate data and penalty gradients.

Verify the weight updates using the shrink factor. Recompute the full objective after the update.

WORKED ANSWER

Answer: prediction and both loss terms

All starting values belong to the fresh practice problem.

$$\hat{y} = 2(1) + 1(2) - 1 = 3, \quad r = 3 - 2 = 1$$

$$L_{data} = \frac{1}{2}(1)^2 = 0.5$$

$$R = \frac{0.1}{2}(2^2 + 1^2) = 0.05(5) = 0.25$$

$$J = 0.5 + 0.25 = 0.75$$

WORKED ANSWER

Answer: every practice gradient and update

Use the same residual 1 in all three data derivatives.

$$g_{w_1} = 1(1) + 0.1(2) = 1 + 0.2 = 1.2$$

$$g_{w_2} = 1(2) + 0.1(1) = 2 + 0.1 = 2.1, \quad g_b = 1$$

$$w_1^{new} = 2 - 0.1(1.2) = 1.88$$

$$w_2^{new} = 1 - 0.1(2.1) = 0.79$$

$$b^{new} = -1 - 0.1(1) = -1.1$$

WORKED ANSWER

Answer: verify the factor and recompute prediction

The factor is .99 because it includes the learning rate.

$$1 - \eta\lambda = 1 - 0.1(0.1) = 0.99$$

$$w_1^{new} = 0.99(2) - 0.1(1) = 1.88$$

$$w_2^{new} = 0.99(1) - 0.1(2) = 0.79$$

$$\hat{y}^{new} = 1.88(1) + 0.79(2) - 1.1 = 2.36$$

$$r^{new} = 2.36 - 2 = 0.36$$

WORKED ANSWER

Answer: the new full objective

The penalty is evaluated at the updated weights too.

$$L_{data}^{new} = \frac{1}{2}(0.36)^2 = 0.0648$$

$$R^{new} = 0.05(1.88^2 + 0.79^2)$$

$$= 0.05(3.5344 + 0.6241) = 0.207925$$

$$J^{new} = 0.0648 + 0.207925 = 0.272725$$

Common errors: penalizing an excluded bias, dividing the regularizer by batch size again, losing the negative-weight sign in its derivative, using $1-\lambda$ instead of $1-\eta\lambda$, or shrinking a second time after already including the penalty gradient.